D07 · Warehouse Operations & WMS

03 · FEFO, lot traceability & quality management in the warehouse

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1 WHAT IT IS

FEFO (first expired, first out) rotates stock by expiry date; lot traceability links every unit to its source lot and destination; warehouse quality management blocks, inspects and releases lots under documented rules.

2 WHY IT MATTERS

For medicines, food and cosmetics, rotation and traceability are consumer protection and regulatory obligations. **FEFO that operators can override is not FEFO, and traceability that is not captured at receiving cannot be reconstructed later.** When a recall happens, lot data decides whether the company withdraws a few lots or an entire product line.

3 HOW IT IS DONE

1. **Capture lot and expiry at receiving.** Mandatory fields on every receipt for regulated and dated products.
2. **Make FEFO a hard system rule.** Lock other locations until the earliest-expiry stock is used.
3. **Integrate warehouse holds with the QMS.** A blocked lot cannot be allocated, picked or shipped.
4. **Run recall simulations.** At least twice a year, timing how fast affected lots are located.

IN PRACTICE

FEFO must be a hard WMS rule — not a suggestion

A WMS that “suggests” the FEFO location but allows the operator to select another location does not guarantee FEFO. The correct implementation locks out other locations until the FEFO location is empty.

Lot traceability requires capturing the lot at receiving — it cannot be added later

The lot and expiry date must be captured at receiving — when the product arrives with its supplier label. Attempting to add lot information after the product has been stored is impractical at scale.

Run recall simulations at least twice a year

A real recall is not the time to discover traceability gaps. Annual (or semi-annual) simulations verify traceability coverage and recall execution time before it’s needed for real.

Integrate warehouse quality management with the company’s QMS

A lot blocked for a quality issue detected in the warehouse must be automatically reflected in the QMS and ERP — so blocked inventory cannot be sold or dispatched while the investigation is managed.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: FEFO implementation with lot traceability — food distributor, 1,200 perishable SKUs — The distributor has 12 expired product complaints at POS per month. Current WMS does not manage FEFO.

Initiative	Implementation	Result
FEFO in the WMS: system only proposes locations with the nearest expiry date	Month 1–2	Expired product complaints at POS: 12/month -> 0 in 3 months post-implementation

Initiative	Implementation	Result
Lot traceability: each receipt captures the supplier lot number and expiry date	Month 2–3	Traceability coverage: 0% -> 94% of receipts with lot captured
Automatic SLOB alert: WMS alerts when a product has <14 days of remaining shelf life in stock	Month 3	First month: 48 near-expiry alerts -> Action: promotions or supplier return

Result: Expired POS complaints: from 12/month to 0 in 3 months. Lot traceability enabled the first recall simulation in 4.2 hours (vs. a typical 8-hour internal target).

HOW IT IS MEASURED

FEFO Compliance % (picks executed following the FEFO rule)

FEFO Compliance %(Picks that took product from the location with the nearest expiry date / Total picks of expiry-date products) × 100

Benchmark: >99% FEFO compliance for pharmaceutical products · >97% for food products FEFO compliance <95% indicates the WMS is not forcing selection of the correct location — or operators are ignoring WMS instructions. Both problems generate expired products at POS.

Lot Traceability Coverage % (% of inventory with full lot traceability)

Lot Traceability Coverage %(Inventory units with lot number and expiry date captured / Total inventory units of expiry-date products) × 100

Benchmark: 100% for pharmaceuticals (regulatory requirement) · >95% for food Complete traceability is the prerequisite of an effective recall. Without lot traceability, a recall requires withdrawing all inventory of a product from all channels — rather than just the affected lots.

WHAT YOU WOULD USE

Traceability & Quality

SAP EWM + QM (Quality Management) — Module: FEFO + Lot Traceability + Quality InspectionNative SAP EWM integration with the Quality Management (QM) module manages FEFO, lot traceability, quality inspections at receiving, and lot blocking for quality issues.

TraceLink / Antares Vision (rfxcel) — Module: Pharmaceutical Serialization + TraceabilityTraceability and serialization platforms for the pharmaceutical industry. Comply with FDA (Drug Supply Chain Security Act) and COFEPRIS traceability requirements (NOM-059-SSA1).

4 SO WHAT

Enforce rotation and capture traceability at the source. Lot and expiry at receiving, FEFO as a hard rule, quality holds the system respects, and recalls rehearsed before they are real.

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